

CONCEPTUAL RESEARCH NOTE | VERSION 0.1.1

Informational Cyclic Cosmology

A Conceptual Hypothesis of Relational Time, Algebra-Relative Entropy, and Aeonically Readability

Patrik William Pustejovsky

ORCID 0009-0008-1618-6619 | Concept DOI 10.5281/zenodo.21115416

Released 2 July 2026 | Open preprint

speculative hypothesis, not a completed physical theory.

ABSTRACT

This note proposes a speculative cosmological hypothesis: fundamental reality may not be spacetime evolving inside an external time parameter, but rather a global informational structure whose internal correlations are locally experienced as time, causality, entropy, matter, and cosmic history. Within this proposal, a cosmic cycle is interpreted not primarily as a temporal restart after the end of the universe, but as a change in the local readability of one globally conserved informational state.

The hypothesis combines ideas from quantum unitarity, black-hole information, holography, thermodynamic coarse-graining, emergent spacetime, relational time, and cyclic cosmology.

No new theorem about Hilbert-space factorization, entanglement relativity, or observational entropy is claimed. The proposed contribution is a speculative cosmological interpretation: an aeonic boundary may correspond to a change in the physically privileged local observable algebra of a stationary global informational state, rather than to a new global state or an externally timed restart. The mechanism selecting such an algebra is not yet provided.

A two-qubit example verifies only a narrower kinematic fact: one fixed pure state can have different bipartite entanglement entropies under different subsystem factorizations. It does not demonstrate a decrease in thermodynamic, observational, or gravitational entropy, nor does it derive a preferred factorization, an aeonic transition, a thermodynamic arrow, or an extension to quantum field theory. The proposal therefore remains a conceptual research program without a distinct quantitative prediction.

Scope and Claimed Contribution

The mathematical ingredients used in this note are drawn from existing work on relational time, observable algebras, factorization-dependent entanglement, quantum mereology, and observational entropy. The finite-dimensional example later in the note is not presented as a new mathematical result.

The narrower claim explored here is that these ingredients may admit a distinct cosmological application. In this interpretation, successive aeons are not fundamentally successive global states. They are different locally readable descriptions associated with different physically privileged observable algebras and clock decompositions of one global state.

This is presently a research question, not an established mechanism. For the proposal to acquire physical content, it must explain what selects the local algebra, in what sense the selection constitutes an aeonic transition rather than a change of notation, and what observable consequence distinguishes it from existing cosmological models.

A non-exhaustive literature review found close prior art for the mathematical ingredients, cosmological observable-algebra embeddings, and relationally induced subsystem algebras. It did not identify the same aeonic interpretation, but that does not establish priority. The claimed contribution is therefore limited to the proposed synthesis and research question.

1. Core Postulates

Foundational postulate: Primacy of the global informational structure

Fundamental reality is hypothesized to be a global informational structure. Time, space, particles, causality, and the thermodynamic arrow are treated as emergent features of how this structure is locally decomposed and read from within.

In the present ansatz, "informational structure" is not an additional substance or a numerical information measure. It is represented provisionally by a Hilbert space, a global observable algebra, a state on that algebra, and a Hamiltonian or constraint. The emergence claims are programmatic and are not derived in this note.

1. Global unitarity

Information is postulated never to be fundamentally destroyed. Apparent information loss is local, coarse-grained, or observer-dependent. In this version, black-hole evaporation must ultimately preserve information.

Because the proposal does not assume a fundamental external time, "global unitarity" does not mean evolution of the universe against an external clock. It means that the fundamental formulation contains no information-destroying non-unitary law and that any relational dynamics recovered within an aeon must be compatible with unitary quantum mechanics.

2. Entropy as loss of local readability

Within the hypothesis, growth of the relevant effective entropy does not mean destruction of the global state. It is interpreted as information becoming increasingly delocalized, correlated, or difficult to reconstruct relative to a specified local algebra and coarse-graining. This is not asserted as a definition of every thermodynamic or information-theoretic entropy.

3. Cyclicity as a property of reading, not temporal restart

The "end" and "beginning" of the universe are not necessarily two events separated by external time. They may be two candidate boundary descriptions of the same global informational structure. What appears as maximal disorder in one local decomposition may correspond to a differently embedded or refactored observable algebra for which a specified effective entropy is low.

4. Low effective entropy through changed distinguishability

The low entropy of a new cosmic phase is not produced by erasing information. It would require a differently embedded or refactored local observable algebra whose induced restricted state has low entropy under a specified entropy functional. Merely discarding observables or coarse-graining more strongly does not guarantee lower entropy and commonly increases observational entropy. This is therefore not the claim that many microstates merely look identical under fixed macro-observables, which would normally correspond to high Boltzmann entropy. The required claim is narrower: the physically selected subsystem structure changes, and the same global state has high effective entropy in one embedded algebra but low effective entropy in the other. The selection and the entropy decrease are not derived here.

5. Cyclicity as a repeatability requirement

A single association between end-of-aeon and beginning-of-aeon descriptions is only a boundary proposal, not yet a cyclic cosmology. Calling the completed model cyclic requires an intrinsically selected, repeatable family of such associations together with conditional histories that run from a low-effective-entropy boundary toward a high-effective-entropy regime within each aeon. This does not require exact periodic recurrence of the same local state or algebra. Repeatability is a requirement of the hypothesis, not a result derived here. A fixed selection rule evaluated on identical global and relational input data would select the same physical algebra again; a sequence of distinct aeonic descriptions therefore requires intrinsically defined relational data that differ between the relevant conditional regimes.

2. Speculative Cosmological Extensions

The following conjectures are motivated by the informational framework but are not logical requirements of the aeonic boundary proposal. Each may be modified or falsified independently without by itself falsifying the central claim about local observable algebras.

2.1 Informational dark-sector conjecture

In this extension, the phenomena conventionally attributed to dark matter are conjectured not to arise from an additional particle species. They are instead the gravitationally accessible signature of correlations retained in the global informational state but inaccessible to the observable algebra of ordinary matter. Those correlations may encode structure associated with matter-like degrees of freedom in another aeonic decomposition while appearing in the current description only through gravity.

Because the central proposal has no fundamental external time, this should not yet be described as a signal propagating from a temporally earlier universe. It is instead a candidate residual pattern in the global state. Calling that pattern a wave would require a propagation equation, characteristic speed, and physical observables that have not been supplied.

At an effective level, one possible implementation would be

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\text{info}}),$$

with total covariant conservation,

$$\nabla^\mu (T_{\mu\nu}^{\text{visible}} + T_{\mu\nu}^{\text{info}}) = 0.$$

Here $T_{\mu\nu}^{\text{info}}$ is only a placeholder. Calling it "information" does not explain why it gravitates. A completed extension must derive it from the global state and selected observable algebra, or replace it with a precise modification of the gravitational field equations. It must also reproduce the empirical successes normally attributed to cold dark matter.

2.2 Emergent rest-mass and scale conjecture

The algebraic boundary ansatz does not by itself require all massive particles to decay. The fate of rest mass becomes compulsory only if the aeonic identification also requires conformal scale invariance, as in conformal cyclic constructions. In that case, the extension must specify whether massive degrees of freedom disappear dynamically or become locally inaccessible under the selected new algebra while their information remains global.

2.3 Dark energy as possible boundary data

Dark energy may drive extreme dilution and thereby contribute to the intrinsic boundary data λ used by the algebra selector. A dynamical dark-energy component might instead participate directly in selecting a new local description. Neither possibility is currently derived. If the cosmological constant remains positive and fixed, the extension must explain why an asymptotic de Sitter regime produces an aeonic algebra change rather than only continued expansion.

3. Algebraic Boundary Ansatz and Missing Selection Rule

For the present type-I ansatz, let $\mathcal{H}$ be a Hilbert space, $\mathcal{M} = \mathcal{B}(\mathcal{H})$ the global observable algebra, H a Hamiltonian or global constraint, and $\rho = |\Psi\rangle\langle\Psi|$ a pure stationary global state. In a Hamiltonian description stationarity means $[H, \rho] = 0$; in a constrained description it may instead take the form $H|\Psi\rangle = 0$. The expectation functional is

$$\omega(X) = \text{Tr}(\rho X), \quad X \in \mathcal{M}.$$

Purity and stationarity are assumptions of this version; neither follows from unitarity alone. The proposed aeonic identification is not represented by replacing one global state with another:

$$|\Psi_{\text{old}}\rangle \rightarrow |\Psi_{\text{new}}\rangle$$

At the schematic level, it instead associates two embedded local descriptions of the same state:

$$(\rho, \iota_{\text{old}}(\mathcal{A}_{\text{old}})) \mathcal{R} (\rho, \iota_{\text{new}}(\mathcal{A}_{\text{new}}))$$

Here:

```
rho = assumed pure, stationary global state
A_old = abstract local observable algebra in the old aeon
A_new = abstract local observable algebra in the new aeon
iota = embedding of an abstract local algebra into the global algebra
```

The proposal concerns the embedding and factorization of local observables inside the same global algebra, not a replacement of the underlying state. Two local algebras may be abstractly isomorphic while being embedded differently in the global algebra and therefore carrying different restricted states. The $\mathcal{B}(\mathcal{H})$ notation is schematic and type-I; it is not asserted to describe local algebras in continuum QFT.

At the present stage, let $\mathcal{R}$ be an unspecified relation contained in the product of the relevant embedding spaces. The symbol T_n labels one candidate pair in that relation:

$$\iota_n^\infty : \mathcal{A}_n^\infty \hookrightarrow \mathcal{M}, \quad \iota_{n+1}^0 : \mathcal{A}_{n+1}^0 \hookrightarrow \mathcal{M},$$

$$T_n := (\iota_n^\infty, \iota_{n+1}^0) \in \mathcal{R}.$$

Here:

A_n^∞ = candidate algebra assigned to the end-boundary regime of aeon n
 A_{n+1}^0 = candidate algebra assigned to the start-boundary regime of aeon $n+1$
 $M = B(H)$ = schematic type-I global algebra
 ι = embedding of a local algebra into the global algebra
 R = proposed boundary relation, not yet a physical law
 T_n = one candidate pair of related embeddings

The superscripts ∞ and 0 are mnemonic boundary labels in this version. They do not denote values of an external time parameter or limits already defined in a topology on states, algebras, or embeddings. If a completed model interprets either boundary as a genuine asymptotic limit, it must specify an intrinsic relational parameter, the relevant topology, and a convergence criterion. The same qualification applies to λ_n^∞ and λ_{n+1}^0 below.

This notation does not define T_n as a unitary operator on $\mathcal{H}$, a completely positive trace-preserving map on states, a homomorphism from the old algebra to the new one, or evolution in an external time parameter. It records the proposed aeonic association and nothing more. The index n labels the candidate aeonic descriptions; it is not an external time coordinate. A full theory must replace $\mathcal{R}$ with an intrinsic rule that selects the two embeddings and relates their operational meanings.

One candidate pair T_n describes only one proposed boundary. A cyclic model would require an intrinsically selected family

$$\{T_n\}_{n \in \mathcal{I}}, \quad T_n = (\iota_n^\infty, \iota_{n+1}^0) \in \mathcal{R},$$

for an aeonic index set $\mathcal{I}$, plus compatible relational histories within the aeons. The index remains bookkeeping for relationally ordered descriptions, not a fundamental external time. Neither the existence nor the repeatability of this family is established here.

The global state is assumed to be the same on both sides:

$$\rho_{n+1} = \rho_n = \rho.$$

For any embedding $\iota : \mathcal{A} \hookrightarrow \mathcal{M}$, the induced state on the abstract local algebra is

$$\omega_\iota(a) := \omega(\iota(a)), \quad a \in \mathcal{A}.$$

Because ρ is assumed pure, its global von Neumann entropy is

$$S_{\text{vN}}(\rho) = 0.$$

The hypothesis requires, rather than derives, a target inequality of the form

$$S_{\text{eff}}(\omega_{\iota_n^\infty}; \mathcal{A}_n^\infty) \gg S_{\text{eff}}(\omega_{\iota_{n+1}^0}; \mathcal{A}_{n+1}^0).$$

Thus the beginning of a new aeon is not caused by information erasure. It is hypothesized to correspond to a change in which distinctions are locally readable. The equations above do not explain why that change occurs or establish the inequality.

Here S_{eff} should not be read as ordinary global von Neumann entropy, which is invariant under unitary evolution. It is a placeholder for an observational or subalgebra-relative entropy and must be specified together with the embedded algebra and coarse-graining. In a finite-dimensional bipartite matrix factor, the restricted state can be represented by a reduced density matrix. For a pure global state, its von Neumann entropy is the bipartite entanglement entropy S_{ent} used in Section 5; it is not automatically the cosmological S_{eff} . More general subalgebras, algebras with centers, and type-III algebras require different constructions. A specified quantum channel could model an operational coarse-graining only after its input algebra, output algebra, and physical meaning are fixed; an arbitrary channel is not a definition of local restriction. None of these entropy constructions defines the relation $\mathcal{R}$ or selects a candidate pair T_n . The current version does not choose a unique entropy construction for the full cosmological case.

Required form of a physical selection rule

Let $\mathfrak{E}_\lambda$ denote a physically admissible family of embeddings of candidate local algebras into $\mathcal{M}$, and let $\sim_{\text{phys}}$ identify embeddings that represent the same operational physics. The label λ must be intrinsic: for example, relational clock data, asymptotic observables, or boundary conditions. It must not be an external time coordinate introduced only to make the aeons consecutive.

A completed model would need a selector of schematic form:

$$[\iota_\star(\lambda)] \in \arg \min_{[\iota] \in \mathfrak{E}_\lambda / \sim_{\text{phys}}} \mathcal{F}([\iota]; \rho, H, \lambda),$$

where H may be a Hamiltonian or global constraint and $\mathcal{F}$ is an operationally defined criterion that must be invariant under $\sim_{\text{phys}}$. Existing quantum-mereology programs suggest criteria based on quasi-classicality, limited entanglement growth, pointer-state stability, locality, or minimal scrambling. The selector must not be defined merely by choosing whichever factorization makes ρ low entropy; that would build the desired conclusion into the premise.

For fixed ρ , H , and λ , a selector with a unique physical minimum would return the same equivalence class of embeddings. Distinct aeonic descriptions therefore require a family of intrinsic relational data, such as λ_n^∞ and λ_{n+1}^0 , for which schematically

$$[\iota_n^\infty] = [\iota_\star(\lambda_n^\infty)], \quad [\iota_{n+1}^0] = [\iota_\star(\lambda_{n+1}^0)].$$

These data must be derived from relational observables, records, conditional states, or boundary structure already encoded in the global theory. They cannot be supplied by the aeon index n , an external time variable, or the desired low-entropy conclusion. Because the clock decomposition is itself selected, a completed model may need a joint consistency condition for the embedding, clock, and relational boundary data rather than treating

λ as an independent input. No such derivation or joint condition is provided here. If the selector instead has degenerate physical minima, the theory must derive a branch-selection or consistency rule; degeneracy alone does not generate an ordered aeonic sequence.

For physical content, the selector must identify a unique or physically equivalent class of embeddings, remain stable under small perturbations, and distinguish operationally meaningful observables from global unitary relabelings. The hypothesis does not yet supply such a functional or prove that its end-of-aeon and beginning-of-aeon solutions exist.

In compact form:

```
same global state rho
+ old embedded algebra -> high specified effective entropy
+ new embedded algebra -> low specified effective entropy
```

4. Relational Clock Time, the Thermodynamic Arrow, and Clock Ambiguity

The Page-Wootters mechanism addresses how conditional dynamics may be encoded in a globally stationary state. It does not by itself explain why one temporal orientation is thermodynamically distinguished. Relational clock time and the thermodynamic arrow are therefore treated as separate problems in this note.

Idealized conditional dynamics

In the idealized construction, the accessible Hilbert space and observable algebra admit compatible clock/rest decompositions:

$$\mathcal{H} \simeq \mathcal{H}_C \otimes \mathcal{H}_R, \quad \mathcal{A}_n \simeq \mathcal{A}_n^C \otimes \mathcal{A}_n^R.$$

The stationary global state must also satisfy an appropriate constraint. In the ideal non-interacting form,

$$H_{\text{tot}}|\Psi\rangle = 0, \quad H_{\text{tot}} = H_C \otimes I_R + I_C \otimes H_R.$$

The zero eigenvalue is conventional; a stationary energy eigenvalue can be shifted to zero. A nonzero constrained state also requires compatible clock and rest energy support, so that contributing energy eigenvalues can sum to the chosen total eigenvalue. Interactions, imperfect clocks, and more general constraints require extensions of this ideal form.

Let the clock states be generated by H_C ,

$$|t\rangle_C = e^{-iH_C t} |0\rangle_C, \quad E_t = |t\rangle_C \langle t|.$$

For $\rho = |\Psi\rangle\langle\Psi|$, the state of the remaining degrees of freedom relative to the clock reading t is

$$\rho_R(t) = \frac{\text{Tr}_C[(E_t \otimes I_R)\rho]}{p(t)}, \quad p(t) = \text{Tr}[(E_t \otimes I_R)\rho].$$

The formula applies where $p(t) > 0$. Each E_t is treated as a rank-one positive effect and, for normalized $|t\rangle$, an individual rank-one projection. A continuous family of such effects in a finite-dimensional clock is generally non-orthogonal and is not a projection-valued time observable; a normalized clock POVM requires an appropriate measure and normalization. Actual measurement updates additionally require a specified instrument. The conditional state above is instead a relative-state assignment and does not require an actual measurement of the clock. Under the ideal constraint and spectral-compatibility assumptions above, it satisfies, in units with $\hbar = 1$,

$$\frac{\partial \rho_R(t)}{\partial t} = -i[H_R, \rho_R(t)].$$

Thus t is an internal label for correlations between a selected clock and the rest, not a fundamental external parameter. The construction recovers reversible conditional dynamics; it does not yet select a preferred clock or a direction of time.

The separate arrow-of-time requirement

The ideal conditional evolution above is unitary. If $\rho_R(t)$ is pure, its von Neumann entropy remains zero. A thermodynamic arrow therefore requires additional structure: a physically selected coarse-graining or accessible subalgebra, a low-observational-entropy boundary condition, and conditional dynamics that produce records and typical entropy growth away from that boundary.

The Page-Wootters construction does not by itself identify either orientation of the clock label as the thermodynamic future. A thermodynamic arrow would orient the clock readings in the direction of typical entropy increase away from the selected low-entropy boundary.

Schematically, for a selected observational algebra $\mathcal{A}_R^{\text{obs}}$, an arrow would require behavior such as

$$S_{\text{obs}}(\rho_R(t_2) | \mathcal{A}_R^{\text{obs}}) \gtrsim S_{\text{obs}}(\rho_R(t_1) | \mathcal{A}_R^{\text{obs}}), \quad t_2 > t_1,$$

for typical conditional histories over an appropriate regime. This is not asserted as exact monotonicity: finite systems can fluctuate or recur. Here S_{obs} is a placeholder for a specified observational or coarse-grained entropy, not the global von Neumann entropy. The proposed low-effective-entropy restricted state relative to the new aeon's algebra could serve as a boundary condition only if that algebra is selected non-circularly and its conditional dynamics actually generate the required asymmetry. Refactorization alone does not do so.

Clock ambiguity

A stationary global state generally admits many candidate clock/rest decompositions. Different choices can encode different conditional histories and even different effective dynamical laws. This is the clock ambiguity. It is especially important here because the proposal already allows the physically privileged observable algebra to differ between aeons.

The status of the ambiguity is disputed. Marletto and Vedral argue that fixing the global Hamiltonian and requiring an ideal non-interacting clock can select a unique suitable decomposition. Recent work by Stoica argues that non-interaction alone is insufficient and that the physical meaning of the observables must also be supplied. This note does not assume that the dispute is settled.

The selector introduced in Section 3 must therefore select not only an embedded local algebra, but also a compatible clock subalgebra and rest algebra, and it must relate them to the fixed global Hamiltonian or constraint and to a physical record structure. It must not choose a clock merely because that clock reproduces the desired

history. Across the aeonic boundary relation, a different selected clock may define a different relational time coordinate, but it does not by itself define a new thermodynamic arrow.

5. Finite-Dimensional Illustration

This section illustrates an established kinematic fact: the entanglement and reduced-state entropy assigned to a pure state depend on the chosen subsystem factorization or observable algebra. It does not model cosmology, derive an aeonic transition, or establish a new mathematical result.

The operational role of observable algebras in selecting quantum subsystems was developed by [Zanardi](#). Explicit constructions show that observables can induce a tensor-product structure in which an arbitrary pure state has a chosen Schmidt decomposition ([Harshman & Ranade](#)), and that a state may appear entangled or separable under different factorizations of its observable algebra ([Thirring et al.](#)). The calculation below is a two-qubit specialization of this known factorization dependence, included to make the proposed cosmological interpretation concrete.

For this illustration only, define the bipartite entanglement entropy S_{ent} as the von Neumann entropy of the reduced state obtained from the assumed pure global state. This quantity depends on the chosen tensor-product factorization. It is not identified here with the cosmological S_{eff} of Section 3 or the observational entropy used for the thermodynamic arrow in Section 4.

Let the global Hilbert space be finite-dimensional:

$$\mathcal{H} \simeq \mathbb{C}^4.$$

In the old aeon, choose a two-qubit factorization:

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B, \quad \dim \mathcal{H}_A = \dim \mathcal{H}_B = 2.$$

Let the globally conserved state be a Bell state with respect to this factorization:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|0_A 0_B\rangle + |1_A 1_B\rangle).$$

The old local observable algebra is:

$$\mathcal{A}_n^\infty = \mathcal{B}(\mathcal{H}_A) \otimes I_B.$$

An observer restricted to this algebra traces over the inaccessible subsystem B :

$$\rho_A = \text{Tr}_B(|\Psi\rangle\langle\Psi|) = \frac{I_A}{2}.$$

The entanglement entropy relative to the old factorization is:

$$S_{\text{ent}}(\rho | \mathcal{A}_n^\infty) = S(\rho_A) = -\text{Tr}(\rho_A \log_2 \rho_A) = 1.$$

Thus, relative to the old factorization, the locally accessible state is maximally mixed.

Now choose a second factorization of the same global Hilbert space, intended only to represent the candidate local description assigned to a new aeon:

$$\mathcal{H} = \mathcal{H}_{A'} \otimes \mathcal{H}_{B'}.$$

Because the tensor-product structure is not fixed by the abstract four-dimensional Hilbert space alone, one may choose a new factorization in which the same vector $|\Psi\rangle$ is separable:

$$|\Psi\rangle = |0_{A'}\rangle \otimes |0_{B'}\rangle.$$

Explicitly, one may define a new product basis by setting:

$$|0_{A'}0_{B'}\rangle = \frac{1}{\sqrt{2}}(|0_A0_B\rangle + |1_A1_B\rangle),$$

$$|0_{A'}1_{B'}\rangle = \frac{1}{\sqrt{2}}(|0_A0_B\rangle - |1_A1_B\rangle),$$

$$|1_{A'}0_{B'}\rangle = |0_A1_B\rangle, \quad |1_{A'}1_{B'}\rangle = |1_A0_B\rangle.$$

This defines a valid tensor-product structure $\mathcal{H}_{A'} \otimes \mathcal{H}_{B'}$ on the same abstract four-dimensional Hilbert space, with the original Bell state now appearing as a product state in the primed factorization.

Equivalently, let U be the global unitary that maps the original computational product basis, in order, to the four primed basis vectors above. Then

$$|\Psi\rangle = U|0_A0_B\rangle, \quad \mathcal{A}_{n+1}^0 = U\mathcal{A}_n^\infty U^\dagger.$$

This makes explicit that the two finite-dimensional factorizations are related by a global unitary. The illustration supplies no physical reason to privilege one embedding over the other.

The new local observable algebra is:

$$\mathcal{A}_{n+1}^0 = \mathcal{B}(\mathcal{H}_{A'}) \otimes I_{B'}.$$

The reduced state seen by the new local observer is:

$$\rho_{A'} = \text{Tr}_{B'}(|\Psi\rangle\langle\Psi|) = |0_{A'}\rangle\langle 0_{A'}|.$$

Therefore:

$$S_{\text{ent}}(\rho | \mathcal{A}_{n+1}^0) = S(\rho_{A'}) = 0.$$

The assumed global state has not changed:

$$\rho_{n+1} = \rho_n = \rho.$$

The global von Neumann entropy remains zero throughout:

$$S_{\text{vN}}(\rho) = 0.$$

The two descriptions differ only in their chosen local factorization and accessible observable algebra.

The values above are entanglement entropies of reduced density matrices. They are not calculations of Boltzmann entropy, observational entropy, horizon or gravitational entropy, or the entropy of a cosmological heat-death state. The illustration therefore does not establish the target inequality for S_{eff} in Section 3; it only demonstrates that one factorization-relative entropy can differ for the same pure vector.

The illustration reproduces the following standard factorization dependence:

```
same global pure state
+ old factorization
-> locally maximally mixed state

same global pure state
+ new factorization
-> locally pure state
```

The calculation does not show that a refactorization occurs, nor that either factorization is physically privileged. In finite-dimensional Hilbert space, an arbitrary refactorization can make entanglement entropy appear or disappear by construction. A physical theory must therefore provide a selection principle for the factorization and separately connect the selected algebra to a thermodynamically relevant entropy. One possible route is quantum mereology: the preferred factorization is not arbitrary, but selected by a Hamiltonian, locality, robustness of pointer states, or quasi-classical dynamics. In the cosmological hypothesis, the missing task is to explain why extreme dilution, dark-energy dominance, or a boundary condition at an aeonic transition would naturally select the new algebra $\mathcal{A}_{n+1}^0$.

There is also a field-theoretic limitation. The illustration uses a finite-dimensional type-I Hilbert-space factorization, where partial traces and reduced density matrices are well-defined. Local algebras in nongravitational relativistic quantum field theory are typically type-III von Neumann algebras, where a spatial tensor-product factorization and ordinary reduced density matrices are not available in the same way (Witten). Gravitationally dressed observable algebras need not have that type: a de Sitter static-patch construction yields a type-II₁ algebra with a corresponding entropy (Chandrasekaran et al.), while clock-defined algebras in semiclassical cosmological spacetimes can be type-II_∞ (Chen & Penington). A full extension must therefore identify the physically relevant algebraic class and define locality, entropy, and restriction without relying on naive partial tracing.

6. Relation to Existing Frameworks

LambdaCDM

The standard model remains the observational baseline. Planck 2018 found strong consistency with six-parameter flat LambdaCDM and no compelling evidence for extensions: Planck 2018. This hypothesis differs by treating the cosmic endpoint as informationally continuous with an effective new beginning.

DESI and dark energy

DESI DR2 BAO measurements are well described by flat LambdaCDM on their own. When combined with CMB and supernova data, however, they favor a time-varying dark-energy equation of state at a significance that depends on the supernova sample, with unknown systematics still a possible qualification: DESI DR2.

This matters because the fate of expansion determines whether dilution, boundary formation, or another transition mechanism is possible.

Penrose CCC

Penrose's conformal cyclic cosmology joins the future conformal infinity of one aeon to the conformally stretched Big Bang of the next across a crossover surface: [Meissner & Penrose](#). Its defining transition is conformal-geometric; it does not identify successive aeons as different locally readable observable algebras of one fixed global quantum state. Proposed CMB Hawking-point evidence remains contested. An et al. claim a detection, while Jow and Scott and Bodnia et al. report no statistically significant evidence: [An et al.](#), [Jow & Scott](#), [Bodnia et al.](#).

Exactly periodic quantum cyclic cosmology

Carroll, Diachenko, and Dulani propose an exactly periodic cosmology in a finite-dimensional Hilbert space and discuss quantum mereology as a still preliminary route to a semiclassical interpretation. Their mechanism relies on unitary state evolution, commensurable energy gaps, and a distinguished low-entropy excursion associated with a Crunch/Bounce/Big-Bang phase: [Carroll, Diachenko & Dulani](#).

The present hypothesis shares the use of finite-dimensional quantum mereology as a possible route to emergent subsystem structure, but proposes a different mechanism. It does not identify the cycle primarily with periodic evolution of the global state. Instead, it asks whether an effectively new aeon can arise through a change in the locally accessible observable algebra or factorization of a globally conserved state. Whether that distinction can be made physically meaningful, rather than merely redescribing the same dynamics, is an open question.

Black-hole information

Recent Page-curve and island results support the possibility that black-hole evaporation can preserve information in quantum gravity models: [Almheiri et al.](#), [Raju review](#).

Emergent spacetime and holography

AdS/CFT supplies a concrete gauge/gravity duality, the Ryu-Takayanagi proposal relates boundary entanglement entropy to bulk minimal-surface area, and Van Raamsdonk argues that bulk connectivity is tied to entanglement: [Maldacena](#), [Ryu-Takayanagi](#), [Van Raamsdonk](#). These results motivate information-theoretic approaches to geometry in specific holographic settings; they do not establish the present cosmological proposal.

Relational time and clock ambiguity

The hypothesis is related to Page-Wootters-style relational time, where a stationary constrained state can encode conditional dynamics relative to an internal clock: [Page & Wootters](#), [Moreva et al.](#), [Marletto & Vedral](#). The ideal construction requires a compatible clock/rest decomposition and constraint; interactions or mixed global states can modify the resulting conditional dynamics: [Rijavec](#).

The non-uniqueness of the clock choice can underdetermine the effective history and laws: [Albrecht & Iglesias](#). Whether ideal non-interaction removes this ambiguity is contested by a recent analysis that instead requires physical meaning to be assigned to the observables: [Stoica](#). None of these constructions derives a thermodynamic arrow from relational time alone. An extended finite-dimensional Page-Wootters model obtains an

informational arrow only after adding inaccessible auxiliary degrees of freedom and effective noisy dynamics: [Jesni Shamsul Shaari](#). This provides an example of the additional coarse-graining needed, not a derivation of a macroscopic cosmological arrow from the clock mechanism itself.

Relational and cosmological observable algebras

Quantum-reference-frame work demonstrates that different relational perspectives can induce different subsystem observable algebras and therefore different notions of subsystem locality and entanglement ([Ahmad et al.](#)). This is close kinematic prior art for the present proposal, but it does not identify those perspectives with successive aeons or supply an entropy-reset mechanism.

Observable-algebra embeddings have also been used to extract a cosmological sector from full loop quantum gravity ([Koslowski](#)). More recent constructions use physical clocks or observer dressing to define type-II gravitational observable algebras in de Sitter and other cosmological settings ([Chandrasekaran et al.](#), [Chen & Penington](#)). These results are relevant to the missing definitions of physical accessibility and effective entropy, but none provides the aeonic boundary relation or repeatable selector proposed here.

Factorization-relative entanglement

The finite-dimensional illustration uses the established fact that subsystem structure, entanglement, and reduced-state entropy depend on the observable algebra or tensor-product structure. This operational and algebraic relativity is developed in [Zanardi](#), [Harshman & Ranade](#), and [Thirring et al.](#). The present note does not extend those mathematical results; it asks whether they admit a physically selected aeonic interpretation.

Quantum mereology and observational entropy

Quantum mereology studies how preferred subsystem decompositions might be selected rather than assumed in advance. Carroll and Singh propose selecting quasi-classical factorizations by limiting entanglement growth and internal spreading: [Carroll & Singh](#). Operational quantum mereology instead represents generalized subsystem structures by an observable algebra and its commutant, and uses minimal algebraic scrambling as a dynamical criterion: [Zanardi et al.](#). A related toy-universe model jointly selects Hamiltonian structure, subsystem decomposition, and a low-entropy state through a variational principle: [Shokrian Zini, Brown & Freedman](#).

None of these results supplies the aeonic boundary relation proposed here, but they constrain what a non-arbitrary selector $\mathcal{F}$ could mean. The entropy used in this note is closer in spirit to observational entropy than to global von Neumann entropy: [Safranek, Deutsch & Aguirre](#), [Buscemi, Schindler & Safranek](#). In quantum field theory, entanglement is more naturally treated as a property of local observable algebras rather than only as a property of states on tensor-product factors: [Witten](#).

Constraints on the informational dark-sector extension

The extension's strongest empirical challenge is dark matter phenomenology. Observations such as the Bullet Cluster show gravitational lensing mass separated from the dominant baryonic X-ray gas, strongly favoring some unseen gravitating component: [Clowe, Randall & Markevitch](#). Any informational-gravitational replacement must reproduce galaxy rotation curves, lensing, CMB acoustic peaks, BAO, cluster collisions, and structure growth at least as well as cold dark matter. The CMB spectra provide an independent precision constraint: a replacement must reproduce the successful base-LambdaCDM fit to the acoustic structure, not only late-time galaxy dynamics ([Planck 2018](#)).

7. Failure Criteria and Missing Physical Content

The central proposal's first vulnerability is the algebra-selection rule. In this version, $\mathcal{R}$ is only an unspecified boundary relation and T_n is one candidate pair of embeddings in that relation, not a quantum channel or time-evolution map. The proposal must define the admissible family $\mathfrak{E}_\lambda$, the intrinsic data λ , and a selector $\mathcal{F}$ that chooses a stable physical equivalence class without assuming the desired entropy decrease. It must also explain how relationally derived data such as λ_n^∞ and λ_{n+1}^0 can differ while the global state and constraint remain fixed; otherwise repeated evaluation selects the same algebra and no aeonic change occurs.

The boundary labels are also not yet mathematical limits. Any asymptotic interpretation of the superscripts ∞ and 0 requires a relational parameter, topology, and convergence law; otherwise they remain labels for the two sides of the proposed association.

The second vulnerability is physical equivalence. If the old and new embeddings differ only by a global unitary relabeling, with no intrinsic operational criterion distinguishing them, the proposal has no physical content beyond a change of description.

The third vulnerability is temporal orientation. Page-Wootters conditioning can recover relational dynamics, but the proposal must separately derive a low-entropy boundary, typical observational-entropy growth, and stable records. It must also show that the algebra selector chooses a physically meaningful clock/rest decomposition rather than one engineered to encode the desired history.

The fourth vulnerability is the field-theoretic and gravitational extension. The finite-dimensional illustration uses type-I tensor-product factorization and partial traces. A realistic version must identify the appropriate local or gravitational observable algebras, which may involve type-III algebras in local QFT or type-II algebras after gravitational dressing, and must define effective entropy without relying on the naive finite-dimensional reduced density matrix.

The fifth vulnerability is observational uniqueness. The hypothesis must predict something different from Λ CDM, Penrose CCC, exactly periodic finite-dimensional quantum cyclic cosmology, and existing emergent-gravity models.

The sixth vulnerability is cyclicity itself. One admissible boundary pair does not establish a cycle. The proposal must show that its intrinsic selection rule can generate a repeatable family of compatible aeonic boundaries and within-aeon relational histories without inserting each new low-entropy description by hand.

Extension-specific failure conditions

The informational dark-sector extension fails in its present form if it cannot derive a covariantly consistent gravitational source and reproduce CMB acoustic peaks, BAO, lensing, cluster collisions, halo phenomenology, and structure growth. A direct particle detection that accounts for the dark-matter phenomenology without a residual informational component would falsify the extension's stated non-particle interpretation, not automatically the central aeonic proposal.

Stable massive particles are fatal only to versions that require a conformal loss of scale. They do not by themselves falsify the more general algebraic boundary relation.

The dark-energy extension remains unsupported until a fixed cosmological constant or a dynamical dark-energy field is shown to provide intrinsic boundary data or a selection mechanism for the new observable algebra.

8. Working Summary

This is not yet a complete physical theory. It is a structured hypothesis:

The universe may be one globally conserved informational structure. Time, entropy, matter, and cosmic history are local readings of that structure. A cosmic cycle would not be a restart in external time, but a change in local readability: the same global information that appears maximally disordered at the end of one description may induce a low-effective-entropy state on a differently embedded or refactored local observable algebra at the beginning of another.

The proposed contribution is this aeonic interpretation of a change in the physically privileged local observable algebra. The note does not claim a new factorization theorem, and it does not yet derive the transition from a Hamiltonian, boundary condition, or other dynamical principle.

To become a physical theory, the hypothesis needs:

1. a rigorous definition of the admissible observable algebras and their embeddings, including a relational definition of the boundary regimes and a topology if genuine limits are claimed,
2. a precise relation $\mathcal{R}$ and a criterion selecting the candidate pair T_n ,
3. an intrinsic, non-circular rule that jointly determines the physically privileged algebra and the relational data λ_n that can distinguish successive conditional regimes,
4. a demonstration that the selected construction is repeatable across aeonic boundaries rather than specifying only one pair,
5. a subalgebra-relative or observational definition of effective entropy S_{eff} ,
6. a field-theoretic version that can handle local operator algebras beyond the finite-dimensional illustration,
7. a compatible relational-clock construction and a separate derivation of the thermodynamic arrow,
8. at least one observational prediction that differs from LambdaCDM, Penrose CCC, exactly periodic quantum cyclic cosmology, and emergent-gravity models.

The optional cosmological extensions additionally require a derivation of $T_{\mu\nu}^{\text{info}}$ or another precise gravitational modification, a clear statement of whether conformal scale loss is required, and a mechanism by which dark energy supplies boundary data rather than merely cosmic dilution.

References

- Ahmad, S. A., Galley, T. D., Hoehn, P. A., Lock, M. P. E., and Smith, A. R. H. (2022). Quantum Relativity of Subsystems. *Physical Review Letters* 128, 170401.
- Albrecht, A., and Iglesias, A. (2008). The clock ambiguity and the emergence of physical laws. *Physical Review D* 77, 063506.
- Almheiri, A., Mahajan, R., Maldacena, J., and Zhao, Y. (2020). The Page curve of Hawking radiation from semiclassical geometry. *Journal of High Energy Physics* 2020, 149.
- An, D., Meissner, K. A., Nurowski, P., and Penrose, R. (2018). Apparent evidence for Hawking points in the CMB Sky. arXiv:1808.01740.
- Bodnia, E., Isenbaev, V., Colburn, K., Swearngin, J., and Bouwmeester, D. (2022). The quest for CMB signatures of Conformal Cyclic Cosmology. arXiv:2208.06021.
- Buscemi, F., Schindler, J., and Safranek, D. (2023). Observational entropy, coarse quantum states, and Petz recovery: information-theoretic properties and bounds. *New Journal of Physics* 25, 053002.
- Carroll, S. M., Diachenko, N., and Dulani, S. (2026). Toward a Phenomenologically Acceptable Quantum Cyclic Universe. PhilSci-Archive preprint.
- Carroll, S. M., and Singh, A. (2021). Quantum Mereology: Factorizing Hilbert Space into Subsystems with Quasi-Classical Dynamics. *Physical Review A* 103, 022213.
- Chandrasekaran, V., Longo, R., Penington, G., and Witten, E. (2023). An Algebra of Observables for de Sitter Space. *Journal of High Energy Physics* 2023(02), 082.
- Chen, C.-H., and Penington, G. (2024). A clock is just a way to tell the time: gravitational algebras in cosmological spacetimes. arXiv:2406.02116.
- Clowe, D., Randall, S. W., and Markevitch, M. (2007). Catching a bullet: direct evidence for the existence of dark matter. *Nuclear Physics B - Proceedings Supplements* 173, 28-31.
- DESI Collaboration (2025). DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints. *Physical Review D* 112, 083515.
- Harshman, N. L., and Ranade, K. S. (2011). Observables can be tailored to change the entanglement of any pure state. *Physical Review A* 84, 012303.
- Jow, D. L., and Scott, D. (2020). Re-evaluating evidence for Hawking points in the CMB. *Journal of Cosmology and Astroparticle Physics* 2020(03), 021.
- Koslowski, T. A. (2007). A Cosmological Sector in Loop Quantum Gravity. arXiv:0711.1098.
- Maldacena, J. M. (1998). The Large N Limit of Superconformal Field Theories and Supergravity. *Advances in Theoretical and Mathematical Physics* 2, 231-252.
- Marletto, C., and Vedral, V. (2017). Evolution without evolution, and without ambiguities. *Physical Review D* 95, 043510.
- Meissner, K. A., and Penrose, R. (2025). The Physics of Conformal Cyclic Cosmology. arXiv:2503.24263.
- Moreva, E., et al. (2014). Time from quantum entanglement: an experimental illustration. *Physical Review A* 89, 052122.
- Page, D. N., and Wootters, W. K. (1983). Evolution without evolution: Dynamics described by stationary observables. *Physical Review D* 27, 2885-2892.
- Planck Collaboration (2020). Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics* 641, A6.
- Raju, S. (2020). Lessons from the Information Paradox. arXiv:2012.05770.
- Rijavec, S. (2023). Robustness of the Page-Wootters construction across different pictures, states of the universe and system-clock interactions. *Physical Review D* 108, 063507.
- Ryu, S., and Takayanagi, T. (2006). Holographic Derivation of Entanglement Entropy from AdS/CFT. *Physical Review Letters* 96, 181602.
- Safranek, D., Deutsch, J. M., and Aguirre, A. (2019). Quantum coarse-grained entropy and thermalization in closed systems. *Physical Review A* 99, 012103.
- Shaari, J. S. (2026). Informational arrow of time in an extended two-qubit Page-Wootters model. *Physics Letters A* 577, 131456.
- Shokrian Zini, M., Brown, A. R., and Freedman, M. (2023). The Smallest Interacting Universe. *Journal of High Energy Physics* 2023, 82.
- Stoica, O. C. (2026). The clock ambiguity problem: extended or extinguished?. arXiv:2604.21805.
- Thirring, W., Bertlmann, R. A., Kohler, P., and Narnhofer, H. (2011). Entanglement or separability: The choice of how to factorize the algebra of a density matrix. *European Physical Journal D* 64, 181-196.

- Van Raamsdonk, M. (2010). [Building up spacetime with quantum entanglement](#). *General Relativity and Gravitation* 42, 2323-2329.
- Witten, E. (2018). [Notes on Some Entanglement Properties of Quantum Field Theory](#). *Reviews of Modern Physics* 90, 045003.
- Zanardi, P. (2001). [Virtual Quantum Subsystems](#). *Physical Review Letters* 87, 077901.
- Zanardi, P., Dallas, E., Andreadakis, F., and Lloyd, S. (2024). [Operational Quantum Mereology and Minimal Scrambling](#). *Quantum* 8, 1406.

Conceptual Provenance and AI-Assistance Disclosure

The originating cosmological intuition is the author's. It developed from questions about Maxwell's demon, statistical entropy, heat death, and whether the apparent end and beginning of a cosmic cycle must describe different universes at all. In particular, the author proposed that the universe remains one and the same global informational reality through the putative cycle, while the apparent succession of an end and a beginning reflects how that information is organized or described. The later formulation in terms of local readability and physically privileged observable algebras was developed with LLM assistance. The informational interpretation of the dark sector also originated in the author's brainstorming, but is retained here only as a separate speculative extension.

Multiple large language models were used extensively during the subsequent development of the manuscript. Their contributions included criticism and counterarguments; development and revision of the operator-algebraic notation, the boundary ansatz, relational-time connections, and the finite-dimensional illustration; literature discovery and comparison; manuscript organization; English drafting and rewriting; and preparation of publication metadata. Much of the present wording and formal presentation was generated or revised with LLM assistance. Some AI outputs made incorrect or overstated claims and were removed or qualified during revision.

No independent expert validation is claimed. The author directed the revisions, selected the claims retained in this version, and accepts responsibility for the final text, citations, errors, and omissions. LLMs are not listed as authors because they cannot verify the work, consent to publication, or assume responsibility for it.